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Content Moderation As a Political Issue: The Twitter Discourse Around Trump's Ban

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Content moderation—the regulation of the material that users create and disseminate online—is an important activity for all social media platforms. While routine, this practice raises significant questions linked to democratic accountability and civil liberties. Following the decision of many platforms to ban Donald J. Trump in the aftermath of the attack on the U.S. Capitol in January 2021, content moderation has increasingly become a politically contested issue. This paper studies that process with a focus on the public discourse on Twitter. The analysis includes over 9 million tweets and retweets posted by over 3 million unique users between January 2020 and April 2021. First, the salience of content moderation was driven by left-leaning users, and "Section 230" was the most important topic across the ideological spectrum. Second, stance towards Section 230 was relatively volatile and increasingly polarized. These findings highlight relevant elements of the ongoing process of political contestation surrounding this issue, and provide a descriptive foundation to understand the politics of content moderation.

Keywords: *content moderation, twitter, public discourse*

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On January 6, 2021 thousands of protesters stormed the Capitol Building in Washington, DC, in an attempt to contest the results of the 2020 US presidential election. Days after the event, Twitter took unprecedented steps to permanently ban the then president Donald J. Trump from the platform after he had voiced support for the protesters, citing a "risk of further incitement of violence" (Twitter, 2021). Shortly after that, other platforms including Facebook followed suit (Rosen and Bickert, 2021). This was widely seen as a pivotal moment in how social media companies enforce their own terms of services and rules in general, and how they treat politicians and other public figures in particular (Wiener, 2021). Facebook had previously shied away from banning President Trump's content, for example, citing public interest and newsworthiness, despite him repeatedly breaking the platforms' terms of services.

Platforms have long laid claim to neutrality when it comes to adjudicating the content circulating on their sites (Gillespie, 2018). Facebook/Meta CEO Mark Zuckerberg repeatedly said that the company should not be seen or indeed become an "arbiter of truth".[1] However, content moderation decisions are part and parcel of social media's business operations—and are far from neutral (Gillespie, 2020, 2018; Gerrard and Thornham, 2020). Over time, most social media platforms have had to develop and refine sets of rules and guidelines about what should and should not be allowed on their sites. These encompass everything from nudity to misleading information, harassment and hate speech, as well as plain old spam, and must balance the need for specificity with the fact that they are implemented in diverse markets. Importantly, making and implementing decisions on the kinds of content that is allowed online is not limited to large internet corporations; instead, the question emerges in virtually any context in which user-generated content is permitted. For example, a niche website popular among crafting enthusiasts, known as "the Facebook of knitting", found itself at the center of controversy after it decided to prohibit users from submitting content such as patterns to knit pro-Trump sweatshirts or Confederate flags.[2]

While the term "content moderation" suggests a menial, relatively harmless activity,

[1] Mark Zuckerberg, Facebook Status Update, Facebook (Nov. 18, 2016) (https://perma.cc/AQ2F-S6EM).

[2] The New Yorker, March 22, 2021 (https://www.newyorker.com/magazine/2021/03/29/how-politics-tested-ravelry-and-the-crafting-community).

it is an increasingly contested practice, linked to fundamental political questions such as freedom of expression. Consequently, it has recently been subject to growing scrutiny by legislators and civil society groups, while Facebook set up a quasi-independent para-judicial institution to adjudicate its decisions (Klonick, 2020). The increased salience of content moderation on the political agenda led to calls for policy action, such as revisions of Section 230 of the United States Communications Decency Act, which allows social media companies to freely moderate user-generated content while protecting them from liability. This paper aims to understand the politics of content moderation by focusing on how the issue has been framed by different sets of actors over time. Problem definition is an essential component of any policy process, but it is particularly important in an area such as content moderation, in which no established consensus exists regarding the nature of the problem, nor the appropriateness of potential solutions. Furthermore, no such consensus is in sight, or is even possible, to a greater degree than in many other policy areas. As Douek (2021, 769) argues, "[o]nline speech governance is a wicked problem with unenviable and perhaps impossible trade-offs. There is no end-state of content moderation with stable rules or regulatory forms; it will always be a matter of contestation, iteration, and technological evolution".

Specifically, in this paper we trace this ongoing "issue definition" process by analyzing the discourse surrounding content moderation on Twitter between January 2020 and April 2021, a period leading up to and following Donald Trump's ban, which contributed decisively to politicizing the issue among the public in the US as well as internationally. We collected and classified about 9.5 million tweets and retweets by about 3.5 million unique users. Using this dataset, our analysis puts forward evidence to understand how content moderation has become a politically contested issue. We do so in two steps. First, we consider the overall discourse on content moderation, taking into account the salience of content moderation as an issue and the topics that are discussed in conjunction to it. Salience peaked around important events including, but not limited to, Trump's ban from social media in January 2021, and was driven by left-leaning users. Section 230—a key piece of legislation regulating online speech (Kosseff, 2019; Keller, 2018)—was the most frequent topic within content moderation, discussed by both left-leaning and right-leaning users. Second, we address the discourse on Section 230 more specifically, with a focus on the stance expressed by different groups of users towards keeping or repealing it. Stance was relatively volatile, particularly among right-leaning users, with signs of increased polarization towards Section 230 towards the end of out observation period. This result points to the ongoing politicization of the issue.

Taken together, the aspects of the discourse highlighted by our analysis address important aspects of content moderation as a political issue and lay the ground for more detailed analyses.

Content Moderation as a Political Issue

Content moderation refers to the “systems for writing and enforcing the rules for what social media platforms allow on their services” (Douek, 2021, 768), or the “organized practice of screening user-generated content” (Roberts, 2019, 12). These systems and practices determine whether content violates a site or platform’s terms of service and involve institutional mechanisms to adjudicate and enforce the rules (Grimmelmann, 2015). Other definitions describe content moderation as a means to “structure participation in an online community” (Grimmelmann, 2015, 47) by setting the standards and norms by which users must abide when interacting with each other. Content moderation is generally performed by a combination of trained human moderators and algorithmic systems designed to parse through large quantities of data to identify potentially problematic material (Gorwa et al., 2020). This process can conceptually be disentangled along two dimensions. Firstly, this review can take place at different stages of the content life cycle, meaning that posts are being screened either before (“ex-ante”) or after (“ex-post”) they have been posted online (Klonick, 2018). The second dimension differentiates between a primarily reactive or proactive approach to content moderation. For the most part, moderators review content only after it has been flagged and brought to their attention, either algorithmically or by users themselves (Klonick, 2018). Yet, mounting pressure from governments and public groups in recent years has nudged platforms to adopt a more proactive approach to the detection and removal of content that violates their own private rules or public laws (Keller et al., 2020; Gillespie, 2020).

How platforms set and enforce rules and standards about the use of their sites has become a major global regulatory and political issue. In Europe, France and Germany were among the first to introduce legislation[3] requiring platforms to identify and remove illegal hate speech within imparted time limits under threat of hefty pecuniary penalties—raising concerns that legislation extended beyond its original scope with potentially negative consequences (Human Rights Watch, 2018; Article 19, 2019). In recent years, Twitter and Facebook have seen a surge in direct government requests—especially from India, Turkey, and Pakistan—to take down content by journalists and publishers (Dang and Culliford, 2021), while other governments are considering banning platforms from moderating content altogether (Nicas, 2021). Despite these efforts to regulate companies' practices, however, there is no consensus on how content should be moderated. Instead, solutions are contested, in part because the issues at stake are themselves subject to political contestation.

On the surface, the lack of consensus regarding appropriate regulatory frameworks for content moderation is surprising. The issue can be perceived as largely technical, an engineering problem that can be solved using modern AI approaches, which is a framing that fits well with Silicon Valley's worldview. However, at a deeper level, content moderation is an issue that raises fundamental questions about democratic accountability and civil liberties. Who should wield the power to adjudicate between permissible and prohibited content? Is the ensuing amplification or suppression of voices online a much welcomed objective, or a deplorable side effect? Under what conditions are government oversight and regulation warranted—and is it even feasible? Specifically, some scholars and civil society groups have raised concerns about the opacity and lack of transparency surrounding content take-downs (Gillespie, 2018; Kaye, 2019; Suzor, 2019), and questioned whether platforms have too much power in devising the set of rules and standards that govern online speech. Others lament platforms' tendency to suppress speech beyond what is required by law at the risk of infringing on users' rights to expression (Keller, 2018; York, 2021). A growing scholarship, finally, points out that platforms' governance regimes only reflect and promote a narrow set of values at the expense of others (Klonick, 2018; Hallinan et al., 2021; Leurs and Zimmer, 2017).

Such fundamental questions about content moderation cannot be resolved purely with technical tools, and different answers generally entail distinct policy responses. In other

[3]the Network Enforcement Act and "Avia" law

words, content moderation is a political issue, which has become increasingly contested. How political actors frame this issue, and how successful they are in doing so, matters. A key insight from the agenda setting literature is that not every problem automatically becomes *political* (Baumgartner and Jones, 1993). In other words, "social conditions do not automatically generate policy actions" Baumgartner and Jones (2010, 27). To rise to the political agenda, a given issue must first be construed as politically salient and specific arguments put forward as to how and why it might warrant policy intervention (Gilardi et al., 2021). Therefore, how political actors frame content moderation may impact the kinds of solutions proposed. For example, if content moderation is primarily framed as a violation of free speech, policy-makers might be more hesitant to implement strict regulation on platforms' rules around hate speech, misinformation and sensitive content. If, in contrast, content moderation is understood as an indispensable tool to fight online harms and create a safe environment for users, regulators might instead be inclined to push platforms towards greater moderation, even at the risk of suppressing legitimate voices. These are just two broad ways in which content moderation can be framed, and many more ways may be thinkable as we are just starting to understand the politics of content moderation.

In an initial step to understand how content moderation has become a politically contested issue, this paper studies the Twitter discourse around Trump's ban. We first focus on the broader discourse, considering both the policies regulating online speech in the United States and how content moderation is conducted in practice. For this part of the analysis, the relevant elements of the discourse include in particular the salience of content moderation as an issue and how content moderation is discussed, that is, in conjunction with which topics. For both questions, the analysis considers distributions over time as well as across different types of users. Then, in a second step, we conduct a more fine-grained analysis with a focus on Section 230, considering the stance expressed by different types of users. Taken together, these elements address key aspects of the politicization of content moderation, as expressed in Twitter discourse.

Data and Methods

Data Collection

Using Twitter's academic API and the R package RTwitterV2,[4], we collected tweets posted between January 2020 and April 2021. Given that our focus is on Trump's ban from social media, the observation period covers the full year leading up to the event as well as several months after it. Our search query reflects the emphasis on both the policies of online speech regulation and the practice of content moderation, as discussed in the previous section. It includes exact phrases (such as "user rules" and "online safety"), hashtags (such as #contentmoderation and #section230), and boolean searches (such as "(content OR user OR users OR account) AND (remove OR removed OR removal OR removals OR takedown OR taken down OR block OR blocked OR suspend OR suspended OR ban OR banned)" and "(Trump AND executive AND order AND (social media OR platforms)"). The full list is reported in the online appendix A.1, which also shows that the results unlikely to be sensitive to the specific choice of search terms.

We filtered the tweets produced by this data collection procedure in two steps. First, following Batzdorfer et al. (2022), we excluded all tweets that were not written in English, removed stop words, convert all tokens to lowercase, and replaced mentions and links with proper tokens. Our initial dataset consists of 20,233,200 tweets posted by 6,326,924 unique users.[5] Second, we classified the tweets for relevance using a supervised classifier that achieved an accuracy of 87%. The procedures are described in detail in online appendix B. The final corpus consists of 2,432,015 original tweets, 154,451 quoted retweets, and 6,991,897 retweets by 3,445,376 unique users. Moreover, for additional analyses, we collected newspapers articles via LexisNexis. The procedure is described in detail in Section E.

[4]https://github.com/MaelKubli/RTwitterV2

[5]About one in three tweets in our dataset is an original tweet, while the rest are retweets. Only about half of the users in our dataset posted an original tweet, while the others only retweeted other users (see online appendix A.2).

Communities and Types of Users

To identify various communities of users who posted about content moderation, we constructed the retweet network. We represent each Twitter account as a node and include a directed edge between nodes if at least one of them retweeted the other one. We only consider nodes whose degree is equal to or greater than 8 (k-core = 8). The resulting graph consists of 16,241 nodes and 165,302 edges.[6] We use the *Louvain* community detection algorithm to identify various communities of users in the retweet network. To characterize the detected communities, as a further step we analyze their top hashtags and *n*-grams, their most influential users (according to their centrality measures), and these users' bio descriptions.

Moreover, we classify users in the following categories, using available lists: members of the 116th[7] and 117th US Congress[8], news outlets[9] and political journalists[10], prominent think tanks[11], and individual experts[12] who are active in the area of content moderation. We label all other users as "Others", which include ordinary citizens, celebrities, influencers, organizations, and anyone else who posted on content moderation during our observation period.

Topics

To identify themes in Twitter users' activity on content moderation, we use Latent Dirichlet Allocation (LDA) by estimating probabilistic models of the topic for each tweet (Blei et al., 2003). These models allow us to identify key content moderation sub-topics that various users discussed over time. We train an LDA model on our entire sample of relevant tweets (excluding the retweets). Each individual tweet is considered a random mix over K topics, which are in turn represented as a distribution over one or two word phrases,

[6] To determine the visual layout of this network, we used the Force Atlas 2 layout in Gephi (https://gephi.org/).
[7] https://www.sbh4all.org/wp-content/uploads/2019/04/116th-Congress-Twitter-Handles.pdf
[8] https://triagecancer.org/congressional-social-media
[9] See a list of 220 national news organizations in the appendix of Eady et al. (2020).
[10] See the appendix section of Alizadeh et al. (2020).
[11] https://guides.library.harvard.edu/hks/think_tank_search/US.
[12] See Table A14 in online appendix for a complete list of individuals.

commonly referred to as tokens. Standard LDA approaches begin by choosing the number of topics for a given corpus k. By qualitatively assessing the semantic coherence of topic models estimated for a range of 3 to 100 individual topics, we found that a model where K is set to 6 classifies tweets most sensibly (see Figure A2 in online appendix for the plot of computed coherence scores).[13] Thereafter, the authors and research assistants manually labelled each topic after having reviewed associated top terms and sample tweets. Finally, each tweet was assigned to its highest probability topic according to the LDA model.

Stance

We use a stance classifier to measure whether tweets express support or disapproval towards repealing Section 230. Specifically, we fine-tuned the model put forward by Barbieri et al. (2021), which performed well in our validations (See online appendix C).

Bot Detection

To address the concern that the Twitter discourse may be driven by bots, we classified a random sample of 100,000 user profiles in our dataset using the Bot-O-Meter API v4[14] (Yang et al., 2019). The procedure is explained in detail in online appendix D. Using a relatively loose threshold for bot classification (80%), we estimate that up to about 9.9% of users in our dataset are bots, which produced about 12.4% of tweets. A slightly more stringent threshold (85%) reduces the estimates to 3% for users and 3.6% for tweets. Based on these results, we are confident that our findings are not driven by bots.

Results

We present two sets of results, addressing key elements of the Twitter discourse on content moderation. First, we consider the overall discourse on content moderation and show (a) how the salience of the issue was distributed over time and across types of users, and (b) the topics discussed in conjunction with content moderation, also over time and types of users (Section 3.1). Second, we concentrate on the discourse on Section 230 and

[13] We used the *Gensim* library of Python for this task with its parameters set as chunksize = 10,000, passes = 50, iterations = 100, alpha = asymmetric, and the default values for the rest of parameters.

[14] https://rapidapi.com/OSoMe/api/botometer-pro

document its stance, over time and across communities and types of users (Section 3.2).

The Twitter Discourse on Content Moderation

Salience

Figure 1 shows the distribution of tweets on content moderation over time. The salience of content moderation was very low until Twitter started fact-checking Donald Trump's tweets in June 2020. After that, the issue only received a moderate amount of attention, as can be seen in the number of retweets. The peak was reached, unsurprisingly, when Donald Trump was initially banned from Twitter for incitement of violence in the context of the assault on the Capitol on January 6, 2021. The timeline of relevant news articles published in this time period shows a similar pattern (see online appendix E for details).

Figure 2 highlights the role of different categories of users by displaying their tweeting activity over time. Four observations stand out. First, non-elite users generate most of the political conversation about content moderation. Second, the number of tweets about content moderation increased significantly around major political events such as the Black Lives Matter (BLM) protests, the US election results, and especially around Trump's ban from major social media platforms. Third, before January 2021, Members of Congress drove the online discussion more than news outlets and political journalists, but their activity decreased after Trump's ban. Fourth, relevant think thanks (e.g., Data and Society Institute, Pew Research Center, and Brookings Institution) and individual experts (e.g., Daphne Keller, Joan Donovan, and David Hoffman) were not the largest contributors to the discussion, compared to other actors.[15]

Next, we focus on the salience of content moderation in different communities defined by retweet patterns. The communities are shown in Figure 3, which illustrates the retweet network of Twitter users who posted about content moderation and either retweeted or were

[15] It might be that experts' tweets are retweeted more frequently by other elite groups, which would enhance their influence. However, that is not the case. The percentage of retweets experts received from the groups we consider is distributed as follows: Congressmembers 0.01%, Think Tanks 0.02%, Experts 1.63%, News Outlets and Journalists 0.04%, Others 98.3%.

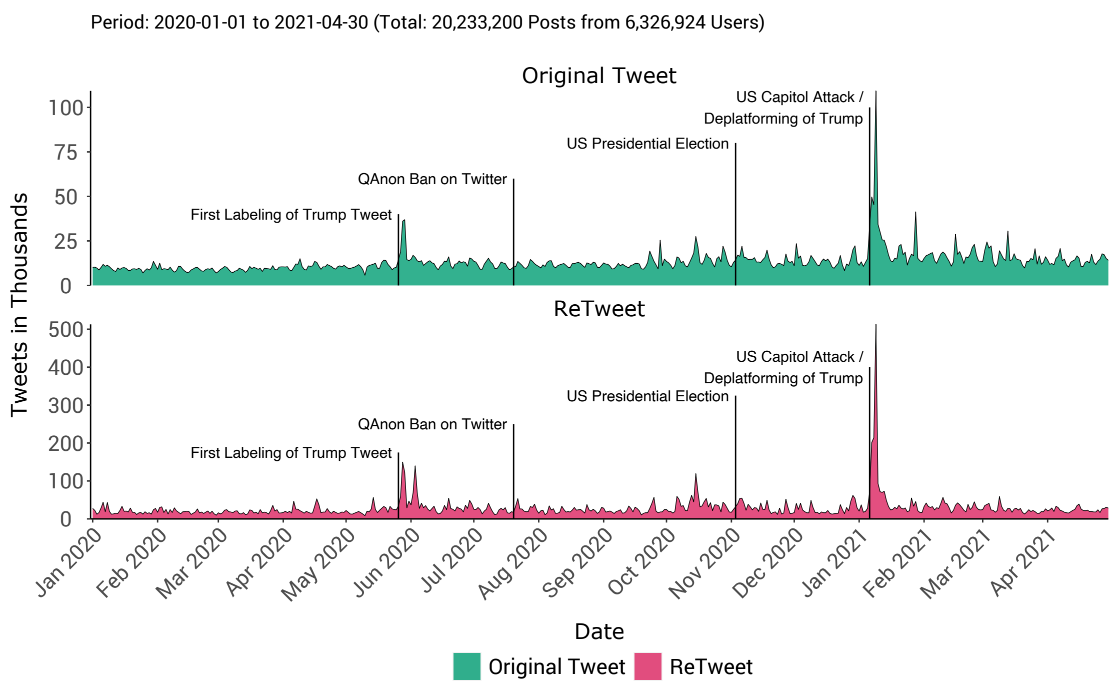


Figure 1. Frequency of original tweets and retweets

Note. Frequency of original tweets and retweets on content moderation. Attention to content moderation peaked when Donald Trump was banned from Twitter for incitement of violence in the context of the assault on the Capitol on January 6, 2021.

retweeted at least 10 times during our data collection period. We can see that the majority of nodes are concentrated in two big, separated clusters, representing left (blue) and right (red) leaning users. More particularly, over 81% of users in the retweet network are clustered in the blue and red communities containing 44% and 37.5% of users respectively (Table 1). This suggests a polarized discourse in which users with similar political affinities retweet each other, but do not retweet out-groups. There are also two smaller communities in Figure 3 which include US and European media accounts (green) and Asian media and users (orange), which contain 9% and 10% of users in the retweet network respectively (Table 1). The most frequently occurring hashtags found in the blue community's tweets are #openinternet, #Section230, #chefsforthepeople, and #twitter. In addition, the most frequent hashtags

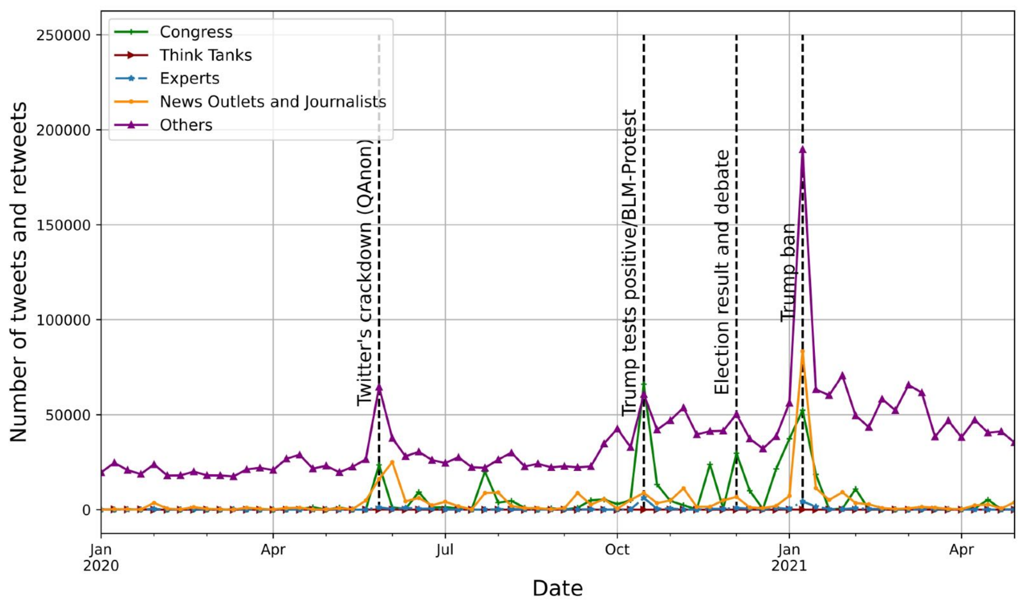


Figure 2. Timeline of the number of tweets and retweets associated with political elites

Note. Plots show the number of tweets produced by each group, plus the number of times they were retweeted by any other account. The majority of the discussion were driven by non-elite users (purple line). Members of the Congress were more active and influential than news outlets and political journalists.

found in the profile descriptions of users in the blue community are #BLM, #Resist, and #BlackLivesMatter. We also see that liberal political commentators and activists such as @kylegriffin1 are among the top accounts with the highest *PageRank* (a measure of identifying influential nodes in a network) score in this community. This evidence leads us to conclude that the blue community leans to the left of the US political spectrum. In contrast, the most frequent hashtags found in the red cluster's tweets are #takeamericaback, #section230, #trump2020, and #2a (referring to the Second Amendment), while the most frequently occurring hashtags in their accounts' description are #MAGA, #KAG (referring to Keep America Great, Trump's 2020 campaign slogan), and #2A. Famous conservative politicians and activists such as @HawleyMO, @RyanAFournier, and @RudyGiuliani are among the top users with the highest *PageRank* score in this community. Hence, we label this community as right-leaning on the U.S. political spectrum. The other smaller

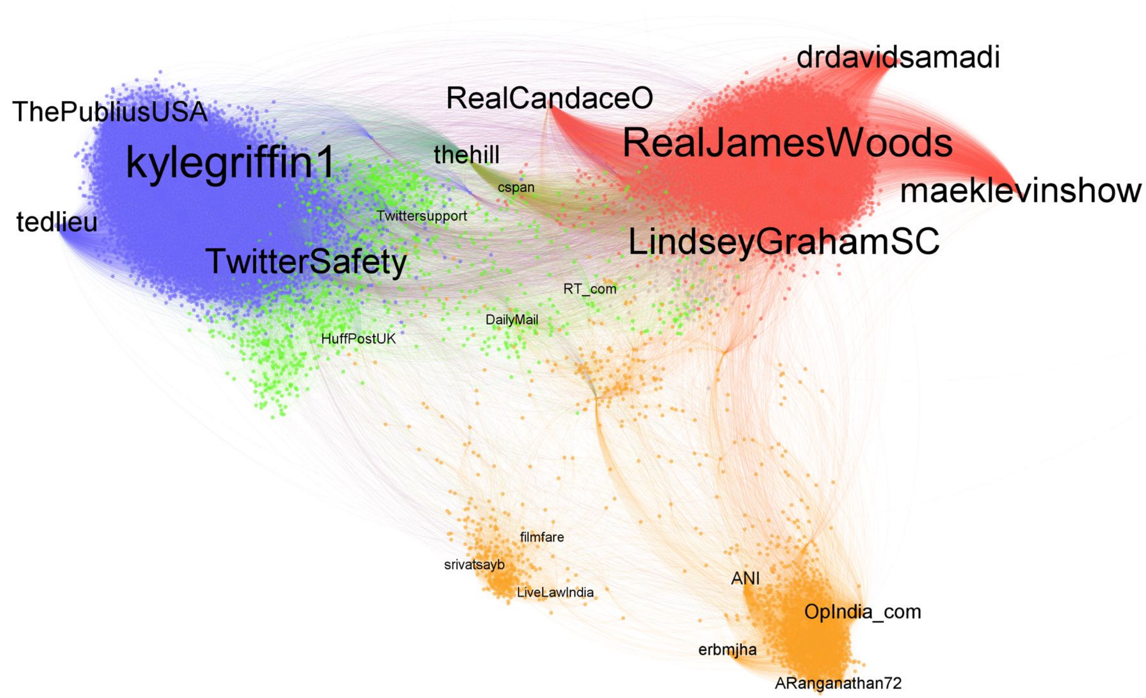


Figure 3. Retweet network

Note. Retweet network of users who posted about content moderation. Each node represents a user, and there is an edge between two nodes if one of them retweeted the other. Node label size represents *PageRank* score. Color reflects different communities identified by the *Louvain* algorithm (Blue: liberal, Red: conservative, Green: US and European media, Orange: Asian Media and Users). Edges are colored by source.

communities are clearly associated with media accounts in Europe, India, and Asia. For a complete list of the key identifying characteristics of the communities, see Section G in online appendix.

Based on these communities, Figure 4 plots the timeline of activity per cluster, where the vertical axis shows the number of tweets originated from users in each community plus the number of times they were retweeted by others. As we have already seen in Figure 1, the distribution of attention to content moderation in all communities peak around major events such as the BLM protests in October 2020, the presidential election and its aftermath events in November and December 2020, and Trump's deplatforming in January 2021. During July 2020, when Trump objected to Twitter's policies, right-leaning users

Table 1: Top words and hashtags

Community	% of Users	% of Tweets	Top Words & Hashtags	Label
Blue	43.9	46.3	#openinternet , #section230, #chefsforthepeople, #twitter, #trumpbanned	Left-leaning
Red	37.4	24.2	#takeamericaback, #section230, #blm, #trump2020, #2a, section, ban, president, trump, repeal	Right-leaning
Green	8.7	18.0	#ban, #partner, #twitchpartner, #twitch, #first-ban, ban, social, Medium, twitch	US & European Media
Orange	10.0	11.6	#whatshappeninginmyanmar, #milkteaalliance, #bringbackpayal, #resignmodi, #apr18coup, terrorist, ban, platform, india,	Asian Media & Users

Note. Top words and hashtags for the communities in identified in the retweet network.

were more active. However, in October 2020, when Trump tested positive for Covid-19 and BLM protests spread throughout the US, left-leaning users were driving much of the activity. Similarly, left-leaning users were more active after Trump had been banned by social media platforms in January 2021.

Topics

Table 2 illustrates the six topics extracted by LDA along with the top 10 words for each topic, description of the topic, and our suggested labels for them. The first topic includes words such as "section", "platform", "speech", and "government", which we label as *Section 230* since most of the tweets are about the repeal or reform of Section 230. The second topic contains top words such as "twitter", "trump", "ban", and "permanantly", which is clearly about social media platforms' ban of Donald Trump. Hence, we label it as *Trump Ban.* Furthermore, "block", "report", and "remove" are among the top 10 words of the third topic, which implies objections to account suspension or removal. Further close-reading of a random sample of tweets associated with topic revealed that most of tweets are personal objections about what happened to their own account, and hence, we label this topic as (*Personal Complaint*). The fourth topic includes top words such as "follow", "rule", "policy", and "twittersafety", which is labelled as *General Complaint* as most of the

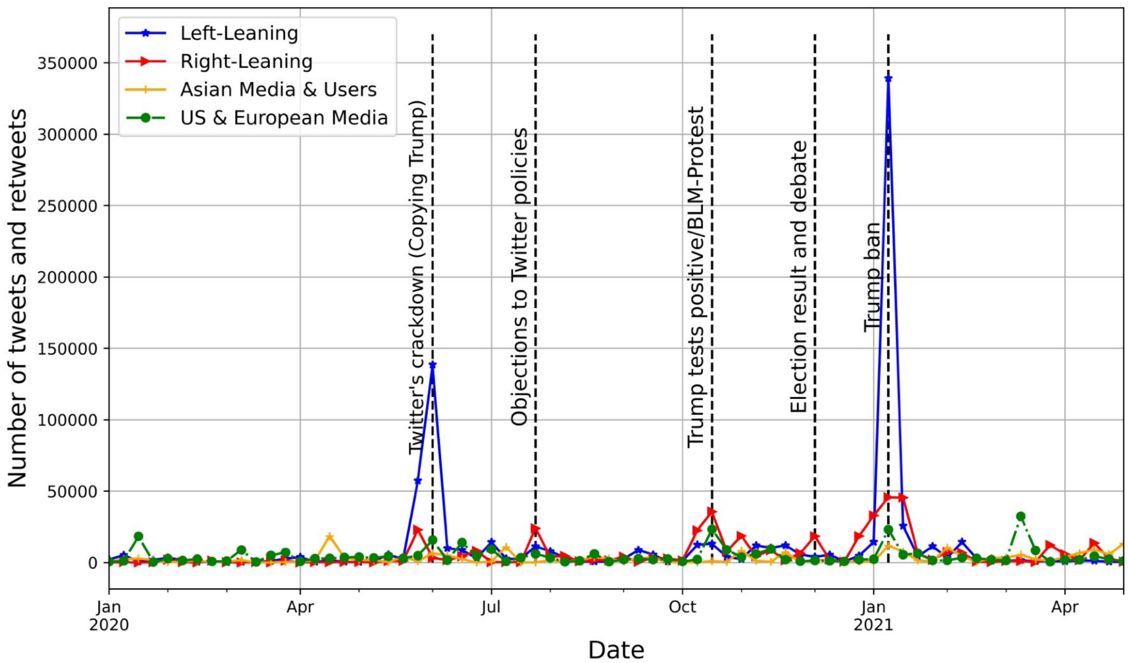


Figure 4. Timeline of the activity of users across different communities

Note. Left-leaning users were driving much of the activity when Twitter labeled Trump's tweets in June 2020 and when Trump was banned by social media platforms in January 2021.

tweets were about objections to Twitter's content moderation actions. In the fifth topic, we see "twittersupport", "reason", and "unsuspend" in the top words. This is suggestive of users' request to *Twitter Support*. Finally, looking at the top words for the sixth topic and reading a sample of 25 random tweets associated with it, it became clear to us that this topic is about social media policies on content moderation, which is why we labelled it as *Platform Policies*. We provide three representative sample tweets for each topic in Table A10 in online appendix.

As can be seen in the last column of Table 2, most of the tweets were about "Section 230" (27% of all relevant tweets and retweets) and "Trump Ban" (21%). Tweets about "Personal Complaint" and "General Complaint" made up 17% and 14% of all tweets respectively, whereas "Twitter Support" and "Platform Policies" comprise 21% of all tweets

Table 2: Results of LDA

No.	Top 10 Words	Description	Label	% of Tweets
1	section, platform, medium, social, donald, trump, speech, content, violence, government, policy, protection	Section 230 Reform/Repeal	Section 230	27%
2	twitter, trump, ban, permanently, realdonaltrump, president, potus, jack, remove, repeal	Platform's Ban of Trump	Trump Ban	21%
3	block, twitter, tweet, report, violence, incite, spread, fake, hate, twitterindia, apple, remove	Objections/Suggestion to Block/Suspend/Report Accounts on Twitter	Personal Complaint	17%
4	follow, rule, policy, twitter, twittersafety, day, follower, tweet, day	Objections to Twitter Rules and Regulation about Approved/Suspended Accounts	General Complaint	14%
5	twitter, twittersupport, rule, violate, suspension, appeal, response, reason, unsuspend, twittersafety	Users' Request to TwitterSuppport	Twitter Support	11%
6	ban, facebook, incitement, violation, community, service, guideline, review, violate, video, content, Instagram, policy, publisher, youtube	Platforms Actions and Policies on Content Moderation	Platform Policies	10%

Note. Extracted Topics from LDA and their Top Words and Suggested Labels.

combined. However, this does not tell us much about the dynamics of topics over time. Therefore, we plot the weekly aggregated numbers of tweets and retweets associated with each topic in Figure 5. Unsurprisingly, “Trump Ban” peaked in January 2021, around the time he was effectively banned by most social media platforms. Moreover, we notice that “Section 230” has been a one of the most salient topics around important events including, but not limited to, Trump’s ban from social media. We will analyze the discourse on Section 230 more in detail in Section 3.2, with a focus on the stance expressed towards it by users in left-leaning and right-leaning communities.

To better understand the underlying characteristics of the political discourse on content moderation, we consider the distribution of topics across the four major communities identified in our network analysis, and in particular left-leaning and right-leaning users. Figure 6a plots the distribution of topics across the three communities of users. We can see that while all four communities were posting about all six topics (except for

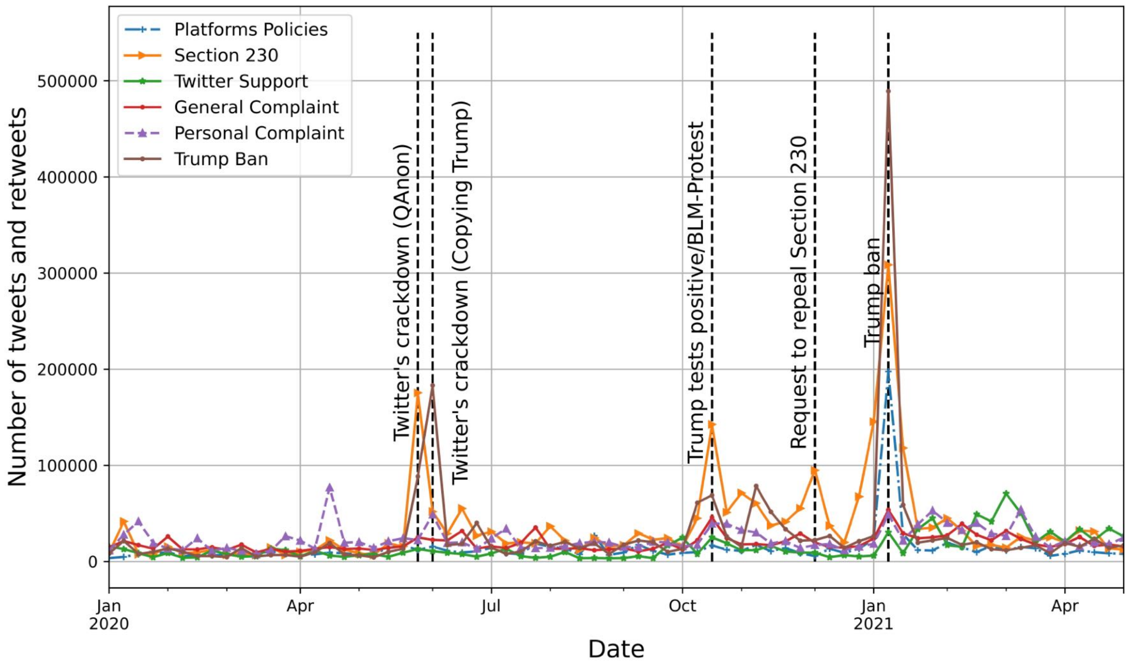


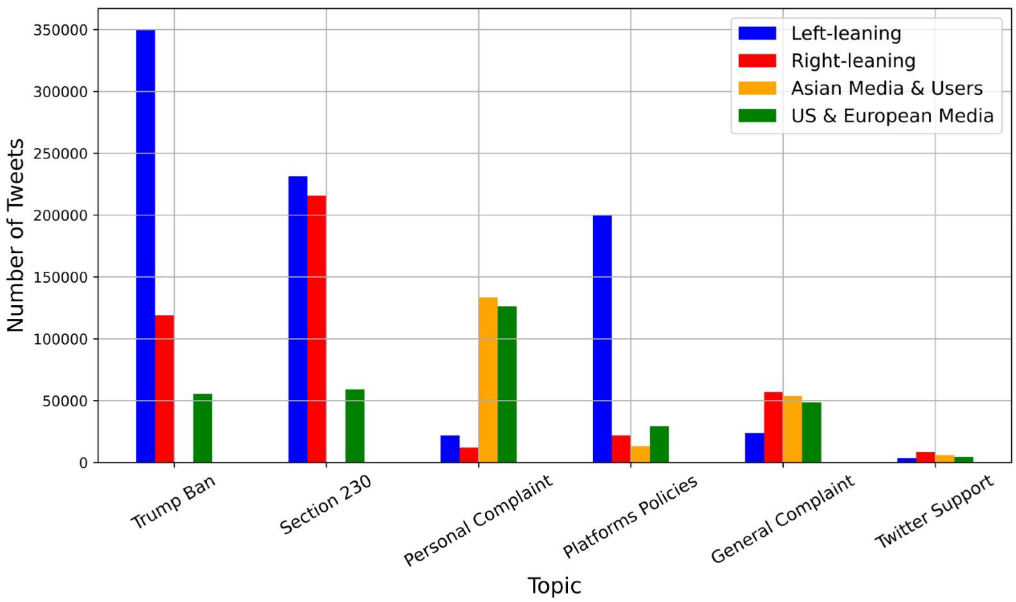
Figure 5. Topic Trends Over Time

Note. ‘Trump Ban” peaked in January 2021, around the time he was banned by most social media platforms. “Section 230” was one of the most salient topics around several important events.

‘Asian Media and Users’ who had not posted about “Trump Ban” and “Section 230”), the amount of activity varies. Users in the left- and right-leaning groups were almost equally posting about “Section 230”. However, whereas much of the discussion about “Trump Ban” and “Platform Policies” were driven by left-leaning users, right-leaning users produced the majority of “General Complaint” and “Twitter Support” posts. These differences highlight the contrasting ways in which users across the political spectrum frame the issue of content moderation. ‘Asian Media and Users’ accounts were mostly posting about “Personal Complaint” and “General Complaint” and had less interest in the other three topics.

Furthermore, 6b displays the share of tweets across the six topics posted by the elite actors identified above (Congress members, think tanks, experts, and news media), that

(a) Distribution of Topics Across Ideological Communities



(b) Distribution of Topics Across Types of Actors

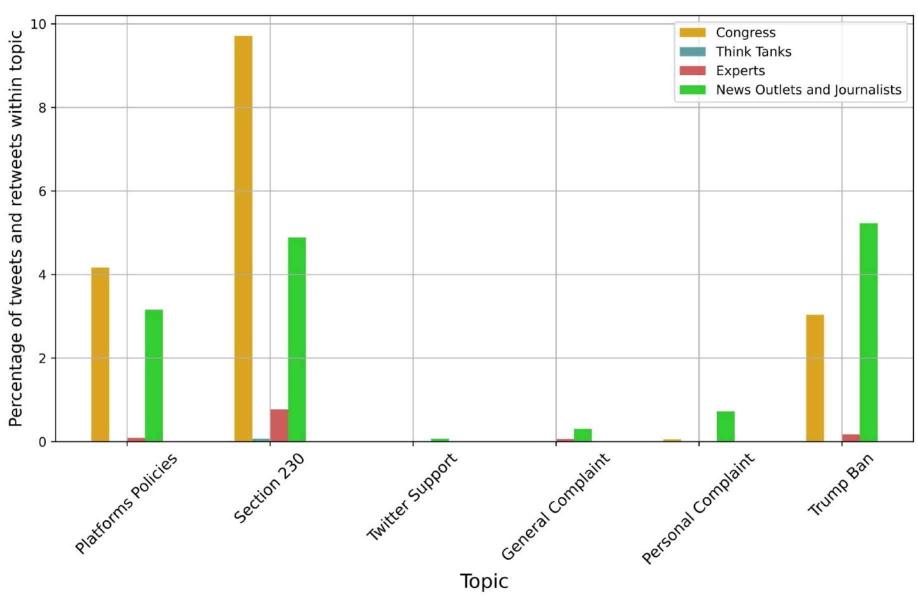


Figure 6. Distribution of topics over communities and user types

Note.(a) Both left- and right-leaning users posted almost equally about 'Section 230'. Much of the discussion about 'Trump Ban' and 'Platform Policies' were driven by left-leaning users, while the discussion around 'General Complaint' and 'Twitter Support' were mostly generated by right-leaning users. (b) Members of the Congress and news outlets and journalists produced the majority of the discussion across all topics. Members of the Congress were more focused on "Section 230" and "Platform Policies".

is, excluding our residual category ("other users"). The first observation is that across all topics, at most 17% of all tweets within a topic were written by members of Congress. For example, we can see that only about 5% of the tweets about "Section 230" were posted by American news outlets and political journalists. The second interesting observation is that members of Congress and news outlets and journalists had the biggest portion of discussion across all topics. The third observation is that members of Congress were much more active than any other group on the "Section 230" and "Platform Policies" topics. Finally, individual experts were much more active and successful in pushing the discussion compared to think tanks.

The Twitter Discourse on Section 230

Communications Decency Act Section 230, passed in 1996, is a key piece of legislation regulating online speech, and the internet more in general (Kosseff, 2019). It protects platforms from liability for content posted by users, while at the same time also protecting them for the content they decide to remove (Keller, 2018, 10). As Figure 5 shows, Section 230 was one of the most important topics in the Twitter discourse throughout our observation period, and particularly surrounding events that increased the salience of content moderation. Moreover, Section 230 was one of the main topics for both Congress members and news media (Figure 6a). Therefore, in this section, we consider the discourse on Section 230 more in detail, with a focus on the stance expressed by users in their tweets—that is, whether they were in favor of repealing Section 230 or repealing it.

Figure 7 shows the distribution of stance on Section 230 during our observation period for the two most important groups of users identified in Figure 3, namely, left-leaning and right-leaning communities. The theoretical range of the stance indicator varies from 0 (in our case, full support for repealing Section 230) to 1 (full support for maintaining Section 230). Several points stand out. First, at the beginning of the period, in January 2020, the average stance of users was similar in both communities, just above 0.5. This result is consistent with the low salience of content moderation at that time, documented in Section 3.1.1. Second, as the salience of content moderation increased following relevant events, stance started to diverge across the right-leaning and left-leaning communities. In the former, stance decreased significantly, consistent with statements by Donald Trump and

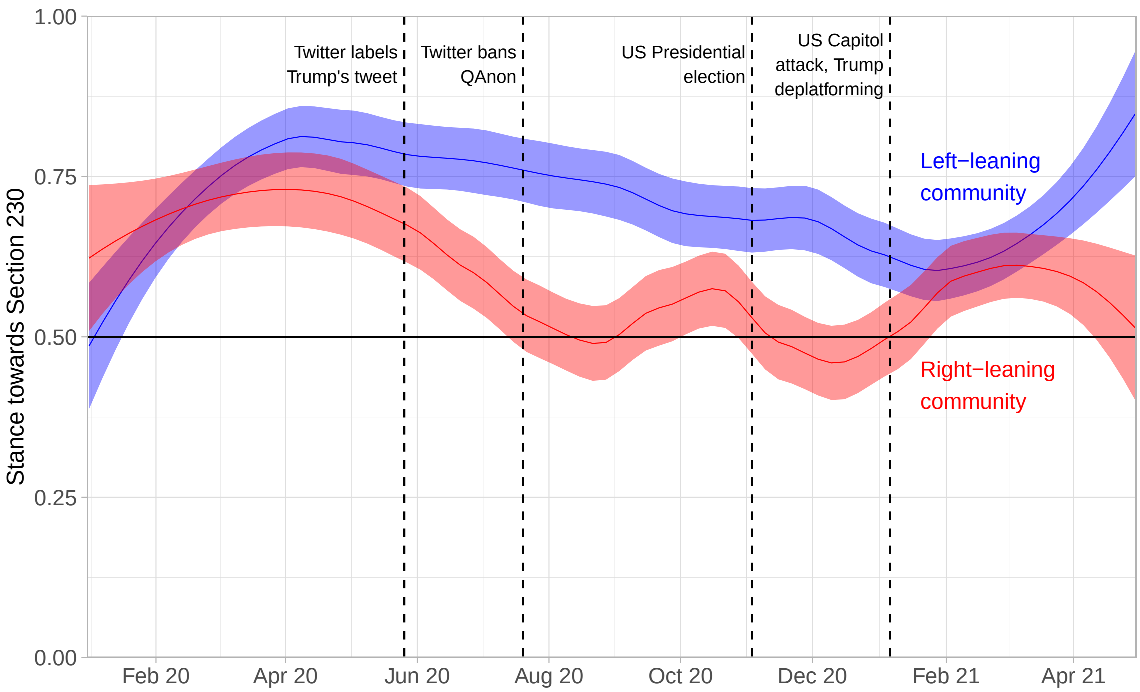


Figure 7. Distribution of stance on Section 230 over time.

Note. 0 means full support for repealing Section 230, while 1 means full support for maintaining Section 230. The changes in stance, particularly in the right-leaning community, points to the ongoing politicization of the issue. The increased polarization visible at the end of the observation period should be checked using longer time series.

other conservative politicians calling for the repeal of Section 230. In the latter, stance also decreased slightly, but remained at a higher level than in the right-leaning group for the most part. Third, stance is relatively volatile, particularly in the right-leaning community. We interpret this fact as a signal of the ongoing politicization of the issue, a process that in 2020-21 was still in progress. Fourth, the divergent trends at the end of the observation period suggest an increasing polarization regarding Section 230. Although a longer time series would be needed to confirm this observation, we interpret the finding as a signal that, by mid-2021, Section 230 may have become subject to degrees of polarization that are typical in most policy areas for the United States.

Conclusion

In this paper, we discussed how content moderation—the "organized practice of screening user-generated content" (Roberts, 2019, 12)—has become a salient political issue in the United States. Our initial analysis of millions of tweets posted between January 2020 and April 2021 shows that public attention to content moderation increased following social media companies' decisions to regulate the activity of prominent political figures. Moreover, the structure of the discourse network, inferred from retweets, reveals a familiar pattern of polarization within US political discourse, with right-leaning and left-leaning users discussing the issue mostly in isolation from each other.

In general, the US Twitter discourse on content moderation revolved around Section 230 (27% of tweets), Donald Trump being banned from social media (21%), specific content moderation decisions by Twitter (17%), general content moderation rules and practices on Twitter (14%), user requests to Twitter support (11%), and platform actions and policies and actions on content moderation (10%). Importantly, different kinds of users engaged with these topics to varying degrees. The number of tweets discussing content moderation from the perspective of "Section 230" was similar among left-leaning and right-leaning users, while left-leaning users posted more tweets about Trump's ban and platform policies than right-leaning users did. Section 230 and platform policies were the two most frequent topics in tweets by Congress members. Regarding Section 230 more specifically, the stance expressed by users in their tweets was relatively volatile with increasing differences between left-leaning and right-leaning users over our observation period, pointing to the ongoing politicization of the issue.

These result highlight relevant elements of the Twitter discourse around content moderation, which point to different ways in which the issue has been politicized in the United States. Although content moderation may be seen as a largely practical or technical problem, it is in fact a process with deep political implications linked to the role of private and public actors in the regulation of speech. How the "problem" is defined matters for the kinds of "solutions" (policy actions) that are advanced to address it.

Our analysis provides an initial basis for a better understanding of the politics of content moderation, which could be extended in a number of ways. First, while the topic models are helpful to understand the general themes associated with content moderation, future research should aim to generate a more accurate conceptualization and measurement of the frames used to describe content moderation as a political problem. Second, different kinds of arenas should be considered, including traditional media and statements made by politicians (for example in Congressional debates) as well as by tech companies. Third, the expansion of the time frame of the analysis would be important, as our findings reflect the specific, relatively limited period we studied. An extension beyond April 2021 would be interesting, particularly regarding stance towards Section 230, to examine whether the increased polarization we observed at the end of our observation period can be confirmed. Fourth, the process by which content moderation has emerged as politically salient issues should be theorized more thoroughly, including the role of different kinds of actors and the connections between different arenas.

Content moderation is a widespread practice which has become increasingly contested due to its significant consequences for democratic accountability and civil liberties. By discussing how content moderation is a relevant political issue (as opposed to a purely technical one) and providing descriptive evidence on the social media discourse around it, we hope that this paper will stimulate further social science research on this important topic.

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Online Appendix

Twitter data

Keywords and stopwords for Twitter API query

Keywords

#deplatforming OR deplatforming OR “user rules” OR #contentmoderation OR #contentremoval OR (content OR user OR account) (remove OR removed OR removal OR removals OR takedown OR “taken down” OR block OR blocked OR suspend OR suspended OR ban OR banned) OR platform OR (social media) (rules OR rule OR policy OR policies OR regulation OR regulations) OR community (moderation OR removal OR removals OR guideline OR guidelines) OR content (moderation OR rule OR rules OR standard OR standards OR policy OR policies) OR ”online safety” OR ”online harm” OR ”online harms” OR ”section 230” OR ”online speech regulation” OR ”platform accountability” OR #onlinesafety OR #section230 OR #onlinespeech OR (trump executive order) (”social media” OR platforms) OR (online speech) (regulation OR regulating OR regulations OR governance OR governing)

Search-term robustness

The construction of our sample is an important step of the analysis, which may be sensitive to the specific search terms that we use. First we separated content moderation into two distinct subcategories, the practice of content moderation and the policies regarding online speech regulation in the United States. We exclude a third category related to the moderation of content itself, as we find it is not possible to come up with a comprehensive list for all the different moderation actions. Therefore, our search terms are as described in the section above, constructed out of our comprehensive table (see Table A1 of search terms.

Table A1: Comprehensive table of all search terms used in the query sorted by sub category

	Practice of content moderation	Policies of online speech regulation
Exact phrases	user rule(s)	online safety online harm(s) section 230 online speech regulation platform accountability
Hashtags	#contentmoderation #contentremoval	#onlinesafety #section230 #onlinespeech
Keywords	(content OR user OR users OR account) AND (remove OR removed OR removal OR removals OR takedown OR taken down OR block OR blocked OR suspend OR suspended OR ban OR banned) (platform OR (social AND media)) AND (rules OR rule OR policy OR policies OR regulation OR regulations) (community AND (moderation OR removal OR removals or guideline OR guidelines)) (content AND (moderation OR rule OR rules OR standard OR standards OR policy OR policies))	(Trump AND executive AND order AND (social media OR platforms) (online AND speech AND (regulation OR regulating OR regulations OR governance OR governing))

Additionally, we consider this potential problem by further checking the robustness of our search string. Specifically, we count how many of all relevant tweets were found by just one of our search term in Table A1. This is a simple but effective procedure to uncover keywords that could have a disproportionate effect on the resulting corpus. Table A2 displays the top 20 terms which would result in the highest reduction in the number of tweets if they were not included in our search string. This represents the drop in results if we would drop one of the search term possibility only. Except for the keywords "account suspended" including its variations and "section 230" and its variations which are central for our research question and clearly should be included in the sample, no other keyword is frequent enough to drive the results on its own. This indicates that the results are not sensitive to the specific choice of keywords.

Table A2: Keywords from the query which generate the most relevant Tweets on their own in the data

Keyword (Search Term)	Number of Hits
account suspend	2,361,552
section 230	939,387
suspended account	913,389
suspend account	729,944
block account	293,375
account remove	279,799
account ban	240,034
account block	189,079
banned account	178,463
account unsuspend	174,015
ban account	158,137
blocked account	142,879
account blocked	131,995
#section230	130,469
remove account	107,279
removed account	101,237
community guideline	82,146
account down	68,494
accounts suspend	64,477
account. suspend	42,495

Original tweets and retweets

Table A3: Retweets vs original tweets

Retweet	Total Tweets	Unique Users
FALSE	6'458'977	2'690'700
TRUE	13'774'223	4'486'809

Table A4: User Statistics for retweets and original tweets

Retweet	Min	Max	Median	Mean
FALSE	1	51'467	1	2.40
TRUE	1	24'432	1	3.07

Classification

For coding purposes, we define content moderation as follows: "Content moderation refers to the practice of screening and monitoring content posted by users on social media sites to determine if the content should be published or not, based on specific rules and guidelines." Using this definition, we instructed research assistants to code tweets as relevant or irrelevant. Relevant tweets include those that discuss social media platforms content moderation rules and practices, and tweets that discuss governments regulation of online content moderation. It also includes tweets that discuss mild forms of content moderation, like flagging tweets and tweets when they directly relate to content moderation. Tweets should moreover be coded as irrelevant if they do not refer to content moderation or if they are themselves examples of moderated content. This includes for example tweets by Donald Trump that Twitter has labeled as disputed a tweet claiming that something is false or a tweet containing sensitive content.

Table A5: Training and Test Data Quality

	Total Tweets	RA 1	RA 2	Average
Test Questions Correctly Classified	365	75.5 %	80.2 %	77.9 %
Classified as relevant	2'559	47.1 %	45.8 %	46.5 %
Classified as relevant and agreed on	2403			43.4 %

To construct the training set, we initially relied on mTurk crowd-workers but then switched to research assistants to achieved the required quality. Our two research assistants classified 2,559 tweets and 365 test questions. Table A5 shows that the agreement with our test questions is fairly good with 77 % and the overall distribution of relevant and irrelevant tweets lies within our expectations. Furthermore the ICC between the two research assistants is very high with 93.9%.

To construct the classifier, we first applied standard text pre-processing steps: we converted all tweet texts to lowercase and removed all URLs, user mentions, and punctuations from the text (we crafted separate features for URLs and user mentions); we removed retweets; and we randomly selected 80% of tweets to train our classifiers on them and report the performance on the remaining unseen out-of-sample 20% of tweets.

Table A6: Classification Performance

	Precision	Recall	F1	Accuracy
Random Forest	0.87	0.86	0.86	0.86
XGBoost	0.87	0.87	0.87	0.87
Logistic Regression	0.85	0.84	0.84	0.85
Linear SVC	0.85	0.84	0.84	0.85

We then calculated five types of human-interpretable features for each post-URL pair: (1) *content*: e.g., word count of a post and URL, polarity and readability of tweets, and *tf-idf* scores for those words that appear in at least 5 tweets; (2) *meta-content*: e.g., top 25 words or bigrams in relevant and irrelevant tweets; (3) *URL domain*: e.g., whether a URL domain is a news, political, satire, or national/local website; (4) *meta URL domain*: e.g., whether a URL domain is in the top 25 political, left, right, national, or local domains shared in relevant and irrelevant tweets; and (5) *communication*: e.g., whether a relevant think tank or congress member has been mentioned in a tweet. In total, we represent each tweet-URL pair as a vector of 3,761 features.

We trained *Random Forests*, *Logistic Regression*, *Linear Support Vector (Linear SVC)* and *XGBoost* classifiers using the *scikit-learn* library for Python Pedregosa et al. (2011). For the *random forests* classifier, we set the number of trees in the forest at 1000 (i.e. *n_estimators = 1000*), and the number of features to consider when looking for the best split as the square root of the number of potential features (i.e. *max_features = sqrt*). We use *scikit-learn*'s default setting for the rest of hyperparameters.

Table A6 reports macro-averaged precision, recall, accuracy, and F1 scores for each class (i.e. relevant or irrelevant to content moderation) using the default classification threshold of 0.5. F1 is the harmonic mean of precision and recall and is a standard metric for binary classification tasks. Precision is the fraction of true positives over the sum of true positives and false positives. Recall is the fraction of true positives over the sum of true positives and false negatives. Our main evaluation metric of interest to choose the best classification model is the accuracy. We achieved the accuracy of 0.87 for the *XGBoost* classifier on 20% unseen out-of-sample test data. Therefore, we choose *XGBoost* as the best model and use it to label the rest of the tweets and exclude the irrelevant ones.

Stance Classification

To make statements about the stance used in regards to section 230 we fine-tuned the transformer model built by Barbieri et al. (2021) which we already use to measure the sentiment of our relevant tweets. To fine-tune this classifier we built a training set with our two research assistants, whom had to code 1,000 tweets. In total we end up with 783 tweets in our training set. Which in turn indicates that the overall ICC of the classification task is 78.3 % between the two research assistants. The training data is relatively well balanced with 329 tweets with a positive stance and 454 tweets with a negative stance (42 % vs. 58 %).

To fine-tune the classifier we first prepossessed the tweets text according to the pre-processing used by (Barbieri et al., 2021). We then trained the pre-trained XLM-T model with the following configurations:

- Data Split: 80 % Training-set; 10 % Validation-set; 10 % Test-set
- Learning Rate: $2e - 5$
- Batch Size: 32
- Number of Epochs: 3

In particular we used a small amount of epochs to prevent over-fitting of the model, due to the relative small training set. We found 3 epochs to be the maximum number which will not result in over-fitting. All other parameters are kept to the specification of the model in Barbieri et al. (2021) paper.

Table A7 reports the macro average, accuracy and the precision, recall and F1 scores for each class. We achieved the accuracy of 0.87 for the fine-tuned stance classification XLM-T model on 10 % unseen out-of-sample test data. Therefore we use this model to classify all tweets related to the topic of section 230 as identified by the topic model to analyze the stance of the topic on Twitter.

Table A7: Classification Performance

	Precision	Recall	F1	Accuracy
Positive Stance	1.0	0.77	0.87	
Negative Stance	0.69	1.0	0.81	
Macro Average	0.85	0.89	0.84	0.85

Table A8: Distribution of stance in 300 random relevant tweets coded by two research assistants and the classifier

Stance by Classifier	Stance by RA 1	Stance by RA 2	Number of Cases	Share of Cases
Contra	Contra	Contra	80	26.9 %
Contra	Contra	Pro	15	5.1 %
Contra	Pro	Contra	10	3.4 %
Contra	Pro	Pro	22	7.4 %
Pro	Contra	Contra	10	3.4 %
Pro	Contra	Pro	5	1.7 %
Pro	Pro	Contra	4	1.4 %
Pro	Pro	Pro	151	50.8 %

To further validate the model we let our two research assistants classify 300 more tweets as either being in favour (pro) section 230 or against (con) it. We compare these results against the labels produced by the classifier by assigning them one of the two classes as well. Table A8 reports the results of this process. Overall the agreement between the research assistants is 88.6 % and between the classifier and both research assistants the agreement is 87.8 % on the cases where both research assistants agreed with each other.

Bot Detection

To get picture regarding the number of bots/automated accounts in the data set. We use the Bot-O-Meter API v4[16] by Yang et al. (2019). This is a machine learning algorithm trained to predict a score where low scores indicate likely human accounts and high scores indicate likely bot accounts. Bot-O-Meter is based on multiple Random Forest classifiers (Breiman, 2001) which produce the score through a large decision tree network. To do this Bot-O-Meter API extracts over a thousand different features to characterize the account's profile, friends, social network structure, temporal activity patterns, language,

[16]https://rapidapi.com/OSoMe/api/botometer-pro

Table A9: Number of user profiles classified as bots

Threshold	Number of Bots	Share of Bots	Content Share by Bots
50 %	59,591	59.6 %	66.6 %
60 %	51,181	51.2 %	58.3 %
70 %	40,849	40.9 %	46.9 %
75 %	33,776	33.8 %	39.1 %
80 %	9,890	9.9 %	12.4 %
85 %	3,035	3.0 %	3.6 %
90 %	739	0.1 %	0.7 %

Note. Number of user profiles classified as bots by the Bot-O-Meter API from a random sample of 100,000 user profiles (99,964 profiles still active) form all 3,45 million user profiles in the relevant data set of tweets.

and sentiment to compute a score. These features are collected through Twitters API and then handed over to Bot-O-Meters API. We classify a random sample of 100,000 User profiles form 3,45 Million user profiles in our relevant Twitter data set.

In Table A9 we show the number of account classified as bots with different threshold values. The threshold might look like a probability score. But in fact according to Yang et al. (2019) the value describes that if an account gets a value of .9, then 90 % of accounts with similar feature values in their training set are bots. At the same time a value of 0.1 would suggest that still 10 % of similar accounts are automated or bots. If we select a threshold of 80 % we can see that the number of bots/automated accounts talking about content moderation is just 9.9 %. If we look at it by the number of tweets the accounts actually posted 12.4 % of all tweets in the sample stem from bots. With a random sample of this size we can make the assumption that these numbers will not differentiate substantially in the full set of tweets. We select a threshold of 80 % simply by the fact that the number of false positives with a lower threshold would be much higher than the actual number of bots in the sample. If we look at the numbers with a threshold of 75 % we would have 25 % false positives according to the workings of the classifier. This is almost as many bots that are classified as such at 80 % including the false positives in this case (9,980 vs. 8,444).

News Articles

We collected data from English newspapers articles collected via LexisNexis[17] API. We gather documents published between January 1st, 2020 and December 31st, 2021. Our search query included several keywords related to content moderation. The full list of queries includes onlinecensorship OR 'online censorship' OR 'corporate censorship' OR deplatforming OR 'section 230' OR 'content moderation' OR contentmoderation OR 'no platforming' OR 'blanket ban' OR 'social media ban' OR 'social media censorship' OR 'content regulation' OR 'moderation tools' OR 'moderation policies' OR 'content filtering' OR shadowbanning OR 'shadow banning' OR 'unverified claims' OR 'disputed claims' OR qanon OR 'trump ban' OR 'misleading information' OR (hateful OR harmful OR extremist OR conservative OR liberal OR objectionable) content OR (trump OR qanon) OR 'Trump ban' OR 'Social Media Censorship' OR 'Content Regulation' OR 'Content Moderation' OR 'Online Censorship' OR 'Section 203' OR 'Social Media Ban' OR 'Misleading Information' OR Deplatforming OR Shadowbanning OR QANON OR Onlinecensorship.

Using this procedure, our initial dataset consists of 976'259 articles, without filtering out the ones that do not contain words related to social media companies or tech companies in particular. If we filter for these articles, we are left with 223,078 articles. Despite our best efforts with the search query and the logic at hand, a significant number of articles were not related to content moderation, and therefore had to be removed further.

To do so, we developed classifiers that achieved accuracy of 0.96 and F1 score of 0.88 using a *XGBoost* classifier with default *scikit-learn* parameter settings. Our final corpus consists of 167,587 articles. The classifier is trained with a training set of 3'374 articles hand coded by two research assistants, with 374 relevant articles and 3,000 irrelevant articles in total[18].

Figure A1 shows the distribution of news articles about content moderation over time. The salience of content moderation increases when Twitter started fact-checking Donald Trump's tweets in June 2020 and he threatened to shut down social media platforms.

[17] https://www.lexisnexis.com/en-us/home.page

[18] Inter coder reliability of the two research assistant is 92.6 %

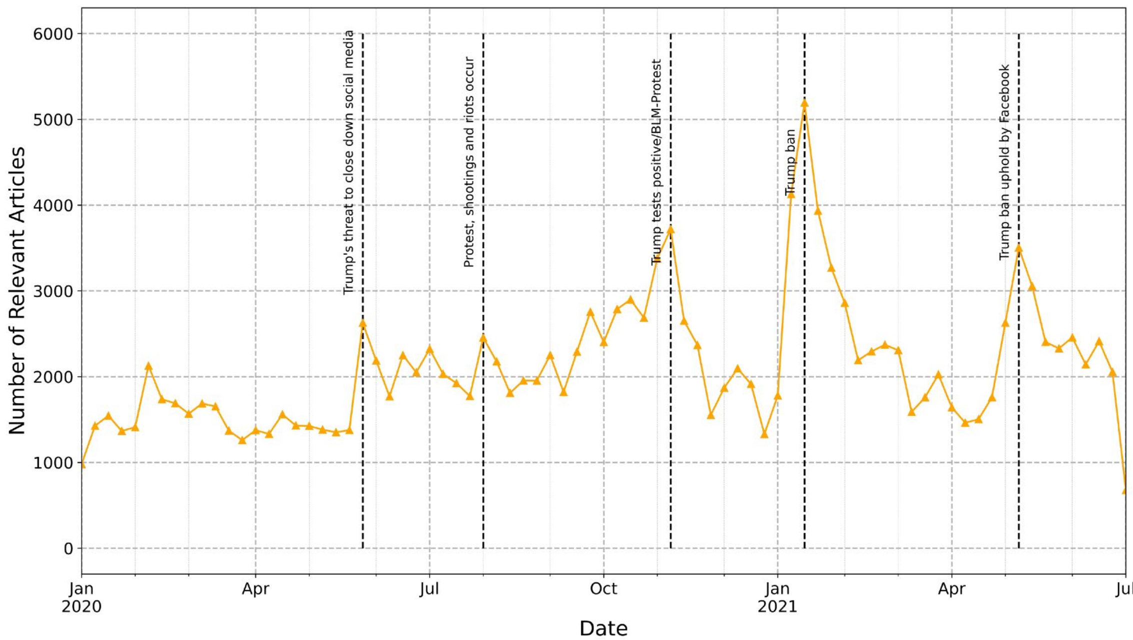

Figure A1. Timeline of News Articles Published about Content Moderation

Note. Data was collected from LexisNexis and classified as relevant or irrelevant using a training data of 3,374 human-annotated articles and a *XGBoost* classifier.

After that, the issue received a relatively higher amount of attention. The peak was reached, unsurprisingly, when Donald Trump was initially banned from Twitter for incitement of violence in the context of the assault on the Capitol on January 2021.

Topic Modeling Evaluation

Figure A2 plots the coherence scores for different number of topics ranging from 3 to 100. Since the coherence score is maximized at K = 6, we picked it as the optimum number of topics.

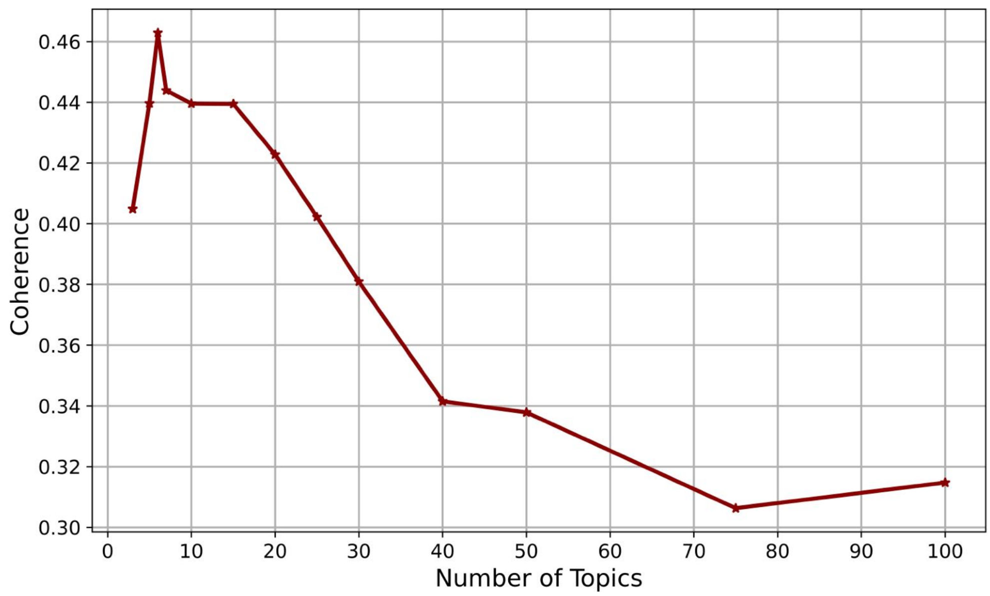

Figure A2. Topic Coherence (Determining Optimal Number of Topics)

Note. We used the *Gensim* library of Python for this task with its parameters set as chunksize = 10,000, passes = 50, iterations = 100, alpha = asymmetric, and the default values for the rest of parameters.

Table A10: Sample Tweets for the Extracted Topics

Topic	Sample Tweets
Platform Policies	- Transparency is fundamental to the work we do at Twitter. Today, we're updating our archive of state-backed information operations with some recent account networks we've removed from our service. - @Facebook strikes again. They are refusing to let @prioritiesUSA run this ad because, according to them, it violates their sensational content policy. "Ads must not contain shocking, sensational, inflammatory, or excessively violent content." Facebook is irreversibly broken. https://t.co/qDFe3tcGPH - How to Create a New YouTube Channel Once Your Account is Suspended? https://t.co/xqK3tqXsu7 #VideoMarketing #videoadvertising #digitalmarketing
Section 230	- Twitter is in violation of Section 230.Unfortunately information to American public is suppressed by the MSM and social media giants. - Donald Trump's misinformation campaigns have left death and destruction in their wake. He's clearly targeting Section 230 because it protects private businesses' right to not have to play host to his lies. - can't believe I have to say this, but..No Section 230 repeal in the NDAA.
Twitter Support	- @TwitterSupport i need help with my account, it has been blocked for no reasons. - Hello @Twitter @TwitterSupport I've been following @1004_linda and I saw this account got suspended for no reason. The owner of this account never violated any of Twitter rules. Please bring back their account ASAP. Thank you!
General Complaint	- Did twitter just suspended account that translating DAY6 content? For real?! I hate this.Is there any conspiration behind all this stuff? Please, protect DAY6 staff too, they're precious for MyDay . - Imagine if Twitter made a wrap up of the year..."Congratulations, you spend 191716161891 minutes on Twitter, engaged in 67 dramas, were insulted 1817 times and canceled twice! " "Your account was suspended once this year "
Personal Complaint	- Please report and block this trash account @/minotazakic for spreading hate and harrassing izone members. - Anuj account was already suspended by @Twitter . If offencive content not shared by Anuj, then why Anuj arrested https://t.co/AtgKGeSFbA. - @realDonaldTrump @Twitter @TwitterSupport these tweets could incite violence. You need to take them down and suspend this account.
Trump Ban	- @realDonaldTrump Enough @Twitter it's time to ban this account or make it so that you can't retweet this garbage. - Trump campaign account suspended after Donald Trump tried to use the account to evade his ban. - JUST IN: Twitter has permanently banned President Trump's account because of "the risk of further incitement of violence" https://t.co/vMrQI7kaEO.

Key Characteristics of the Communities

This section provides key features of the detected communities described in Section 4.2.

Table A11: Characteristics of the Left-Leaning Community

	Top Tweeter	Top Retweeted	Top Influential	Top Shared URLs	Top Hashtags	Top Mentioned	Top Words
1	tsartbot	briantylercohen	briantylercohen	hill.cm	#diaperdon	@briantylercohen	trump
2	GovSherazKhan	SachaBaronCohen	tedlieu	trib.al	#markzuckerberg	@sachabaroncohen	ban
3	Eilonwy24	sociauxling	Acosta	cnn.it	#biglie	@blackmarxist	twitter
4	MaryJan62216031	tedlieu	juliettekayyem	washingtonpost.com	#section230	@jack	president
5	bethlevin	HKrassenstein	kylegriffin1	cnn.com	#breaking	@realdonaldtrump	permanent
6	ewindham3	SenWarren	HKrassenstein	rawstory.com	#executiveorder	@twitter	run
7	givemepie360	Acosta	SachaBaronCohen	newsweek.com	#annomynous	@acosta	break
8	BeHappyandCivil	CNN	clairecmc	politico.com	#annonymous	@tedlieu	section
9	ScoobyLady27	juliettekayyem	CaslerNoel	variety.com	#weirdo	@hkrassenstein	people
10	Elizabe29599604	scottjshapiro	SenWarren	dlvr.it	#weirdotrump	@senwarren	violent
11	PamUnplugged	clairecmc	scottjshapiro	axios.com	#crybabytrump	@cnn	announce
12	SandyKAlano	NPR	AnaCabrera	politi.co	#dontaskdonttell	@juliettekayyem	deplatforming
13	DemocratsUp	CaslerNoel	robertjdenault	kateklonick.com	#consequencesfortrump	@susanwojcicki	fake
14	proudCanadavet	kylegriffin1	JoeBiden	scotsman.com	#donstake	@sundarpichai	medium
15	Tanis42	thehill	CNN	abc.net.au	#tiktokban	@scottjshapiro	facebook
16	ToniRagusa	JoeBiden	TheRickyDavila	thehill.com	#sikh	@clairecmc	social
17	TechmemeChatter	AnaCabrera	MSNBC	nbcnews.com	#trump	@npr	time
18	kat223	MSNBC	JuddLegum	thedailybeast.com	#parlerapp	@caslernoel	donald
19	bassher	Meidas_Michelle	thehill	abc7.la	#bears	@kylegriffin1	understand
20	roripierpont1	David_Leavitt	NormEisen	usatoday.com	#smartnews	@realdonaldtr	celebrity

Table A12: Characteristics of the Right-Leaning Community

	Top Tweeter	Top Retweeted	Top Influential	Top Shared URLs	Top Hashtags	Top Mentioned	Top Words	Top Bigrams
1	real_Stephanie	LindseyGrahamSC	HawleyMO	trib.al	#section230	@realdonaldtrump	section	
2	ldga123456	HawleyMO	LindseyGrahamSC	hann.it	#bigtech	@lindseygrahamsc	ban	
3	PatriotMarie	RyanAFournier	RyanAFournier	childrenshealthdefense.org	#censorship	@hawleymo	trump	
4	MtRushmore2016	RudyGiuliani	RudyGiuliani	ow.ly	#thedefender	@ryanafournier	twitter	
5	Micropan44	GOPLeader	GOPLeader	foxnews.com	#instagram's	@rudygiuliani	president	
6	Samtaylorrose	RealJamesWoods	RealJamesWoods	buff.ly	#sect…	@gopleader	social	
7	StayTheCourse7	GayStr8Shooter	jsolomonReports	justthenews.com	#section230.	@realjameswoods	medium	
8	perchance99	laurenboebert	laurenboebert	justice.gov	#antifa	@gaystr8shooter	repeal	
9	21stcenturycrim	jsolomonReports	GayStr8Shooter	wsj.com	#section23…	@laurenboebert	tech	
10	mike71914654	Jim_Jordan	Jim_Jordan	dlvr.it	#2a…	@jsolomonreports	big	
11	Matthew52802818	MaryVought	RepTedBudd	reut.rs	#section230?	@twitter	censorship	
12	BeanK511	nypost	seanhannity	babylonbee.com	#walkaway	@jim_jordan	bill	
13	M_William_1985	RepTedBudd	MaryVought	truepundit.com	#nsl]	@maryvought	vote	
14	MyPlace4U	seanhannity	nypost	whitehouse.gov	#antielab	@nypost	biden	
15	rollypoly31	TheLeoTerrell	TheLeoTerrell	proofpoint.com	#antifadomesticterrorists	@reptedbudd	fracking	
16	Nmharley2	barronjohn1946	CharlesPHerring	thegreggjarrett.com	#new	@jack	time	
17	OldSalz	CharlesPHerring	SeanLangille	wnd.com	#liberalstudies	@seanhannity	speech	
18	lori_clydesdale	MrAndyNgo	RepGregSteube	peoplesgazette.com	#hongkong	@fdrlst	break	
19	edmgail1944	RepGregSteube	MrAndyNgo	appledaily.com	#foxnews	@theleoterrell	conservative	
20	Draggen75	CawthornforNC	CawthornforNC	hannity.com	#sotu	@barronjohn1946	account	

Table A13: Characteristics of the Foreign Media Community

	Top Tweeter	Top Retweeted	Top Influential	Top Shared URLs	Top Hashtags	Top Mentioned	Top Words	Top Bigrams
1	shadesmaclean	mtracey	mtracey	independent.co.uk	#exclusive:	@mtracey	ban	
2	virgiliocorrado	navalny	navalny	bbc.in	#toolkits,	@navalny	trump	
3	cgsheldon	authoramish	PeterSweden7	theguardian.com	#censorship	@authoramish	twitter	
4	vmrwanda	dhruv__rathee	officialmcafee	zerohedge.com	#farmerspro...	@dhruv__rathee	donald	
5	world__news__eng	PeterSweden7	zerohedge	tdrt.io	#twitter	@petersweden7	medium	
6	Varun8Vijay	spectatorindex	OpIndia__com	patreon.com	#trump	@amitmalviya	censorship	
7	meghnabasu2	OpIndia__com	DailyMail	trib.al	#kooapp...	@spectatorindex	social	
8	newscenterPHL1	amitmalviya	Independent	indy100.com	#dominiccummngs	@opindia__com	content	
9	PHLNewsInsider	TheBrando2	kittypurrzog	opindia.com	#dominiccummings	@thebrando2	speech	
10	TechmemeChatter	BefittingFacts	spectatorindex	tcrn.ch	#breaking	@befittingfacts	free	
11	SouravDindaBJP	NorbertElekes	techdirt	republicworld.com	#narendramodi	@norbertelekes	deplatforming	
12	BitesData	officialmcafee	NorbertElekes	theverge.com	#deplatforming	@twitter	president	
13	varun18vijay	piersmorgan	authoramish	sptnkne.ws	#tommyrobinson	@officialmcafee	platform	
14	VirtualPartyBla	Independent	amitmalviya	punchng.com	#left:	@piersmorgan	account	
15	dhruvbhim	KapilSibal	Snowden	engt.co	#europeanparliament	@independent	amp	
16	allymrowe	saltydkdan	MeghUpdates	ndtv.com	#kashmir	@kapilsibal	facebook	
17	jazmasigan__2	Snowden	SkyNews	dlvr.it	#facebook	@saltydkdan	extend	
18	PhilDeCarolis	zerohedge	coolfunnytshirt	rol.st	#section230	@youtube	act	
19	rmrbq	DailyMail	MichaelChongMP	ctvnews.ca	#update	@joerogan	thread	
20	InnovativeHindu	newslaundry	ellymelly	indiatimes.com	#gravitas	@snowden	unacceptable	

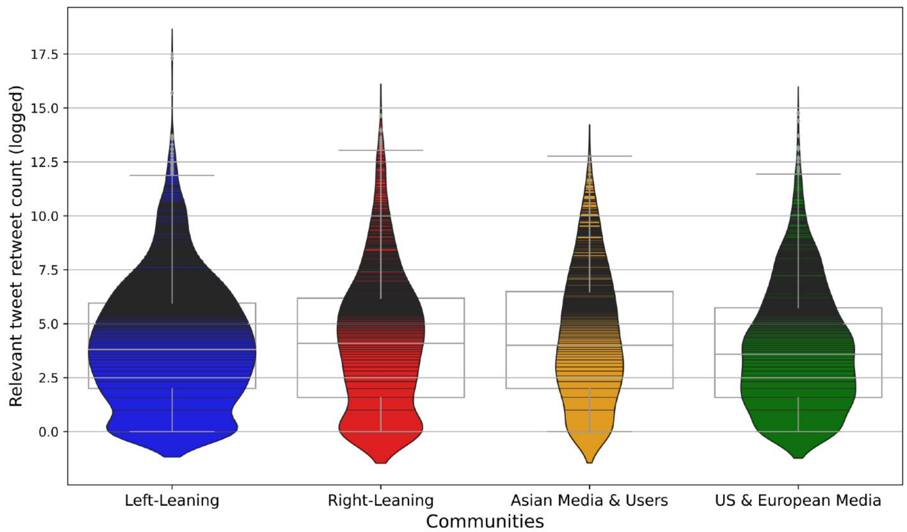

Figure A3. Retweets distribution across three communities

Note. Kernel density plots with overlaid boxplots showing median, first, and third quartiles of the distribution.

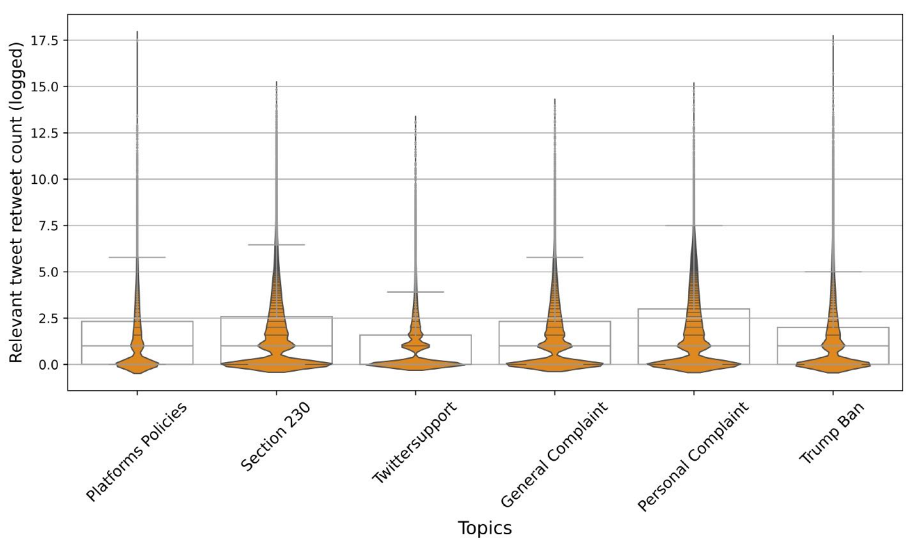

Figure A4. Retweets distribution across six topics

Note. Kernel density plots with overlaid boxplots showing median, first, and third quartiles of the distribution.

Table A14: List of the individual experts on the topic of content moderation and their Twitter handles

Name	Twitter Account
Ajit Pai	@AjitPai
Jessica Rosenworcel	@JRosenworcel
Kelly Quinn	@__kquinn__
Sarah Lamdan	@greenarchives1
Johannes M. Bauer	@jm__bauer
Dr. Ruth, Ph.D.	@LCCWC__Ruth
Jasmine McNealy	@JasmineMcNealy
Jonathan Obar	@CDNJobar
Woodrow Hartzog	@hartzog
Peter Swire	@peterswire
Rich DeMillo	@richde
Shaheen Kanthawala	@ItsShaheenK
Azer Bestavros	@Bestavros
alessandro acquisti	@ssnstudy
Laura Brandimarte	@thefirstred
Philip Napoli	@pmnapoli
Robyn Caplan	@robyncaplan
Jacob Metcalf	@undersequoias
Elizabeth Anne Watkins	@watkins__welcome
Tarleton Gillespie	@TarletonG
danah boyd	@zephoria
Chris Bail	@chris__bail
Daniel Kreiss	@kreissdaniel
Shannon McGregor	@shannimcg
Robyn Caplan	@robyncaplan
Casey Fiesler	@cfiesler
Brianna Dym	@BriannaDym
Michael Zimmer	@michaelzimmer
Aaron Jiang	@aaroniidx
Hannah Bloch-Wehba	@HBWHBWHBW
Nathaniel Raymond	@nattyray11
Sofia Ranchordas	@SRanchordas
Elisabeth Sylvan	@lisard
Joan Donovan	@BostonJoan
Deen Freelon	@dfreelon
Alexandra Siegel	@aasiegel
Kelly Quinn	@kelly___quinn
JaneCombs	@jane__combs
Philip Napoli	@pmnapoli
Ken Rogerson	@rogerson
Seth C. Lewis	@SethCLewis
Sarah Stonbely	@SarahStonbely
Tom Glaisyer	@tglaisyer
colaresi	@colaresi
Jen Easterly	@CISAJen
Daphne Keller	@daphnehk
marius dragomir	@mariusdrag
David Hoffman	@hofftechpolicy
Jolynn Dellinger	@MindingPrivacy
Justin Sherman	@jshermcyber
Francesca Tripodi	@ftripodi

Table A15: Top 10 users in each community who gained the highest retweets and likes

Left-Leaning Screen Name	%	Right-Leaning Screen Name	%	Foreign Media Screen Name	%
briantylercohen	12.8	LindseyGrahamSC	14.5	Jack_Septic_Eye	6.1
rickygervais	7.3	HawleyMO	8.4	mtracey	5.1
SachaBaronCohen	5.9	RyanAFournier	4.9	saltydkdan	3.1
sociauxling	2.7	RudyGiuliani	4.7	navalny	3.1
CNN	2.6	GOPLeader	2.7	jazz_inmypants	2.7
BetteMidler	2.6	laurenboebert	2.6	dhruv_rathee	2.5
scottjshapiro	2.3	RealJamesWoods	2.3	authoramish	2.5
HKrassenstein	2.2	jsolomonReports	2.2	piersmorgan	2.3
kylegriffin1	1.9	barronjohn1946	1.9	NorbertElekes	1.9
SenWarren	1.9	GayStr8Shooter	1.8	ContraPoints	1.7

Table A16: Top 10 users in each community who gained the highest retweets

Left-Leaning Screen Name	%	Right-Leaning Screen Name	%	Foreign Media Screen Name	%
briantylercohen	9.36	LindseyGrahamSC	9.29	mtracey	5.27
SachaBaronCohen	7.18	HawleyMO	9.09	navalny	4.20
sociauxling	4.77	RyanAFournier	5.54	authoramish	3.03
tedlieu	2.58	RudyGiuliani	4.69	dhruv_rathee	2.48
CNN	2.54	GOPLeader	3.48	OpIndia_com	2.36
SenWarren	2.53	RealJamesWoods	2.94	PeterSweden7	1.91
HKrassenstein	2.51	jsolomonReports	2.70	spectatorindex	1.84
Acosta	2.30	laurenboebert	2.60	Independent	1.78
juliettekayyem	2.18	GayStr8Shooter	2.53	BefittingFacts	1.70
scottjshapiro	2.07	Jim_Jordan	2.16	amitmalviya	1.69